

Spin-Dependent Exotic Interactions

Matter–Antimatter Asymmetry Analysis Report

Generated: 2026-05-14 19:44:01

Metric	Value
Pairs analysed	7
Pairs skipped	0
Analysis timestamp	2026-05-14 19:44:01
Framework	SPINDEP v1.0
Method	log-interpolated chi-squared asymmetry
Chi-squared modes	Uniform (10%) + Weighted (per-point curvature)

Summary of All Pairs

Coupling	Potential	Sector	Matter	Antimatter	A mean	χ^2 uniform	χ^2 weighted	p-value (w)
gsgs	UNKNOWN	ee	Delaunay2017	Adkins2022	0.873	93299	10772	0.00e+00 **
gAgA	V1	ep	Karshenboim2011	Fadeev2022	1.000	119956	101909	0.00e+00 **
gAgA	V2	ee	Kotler2015	Fadeev2022	1.000	119990	33391	0.00e+00 **
gAgA	V2	ee	Ficek2017	Fadeev2022	0.999	119759	65969	0.00e+00 **
gAgA	V2	ee	Ficek2017	Fadeev2022	0.954	109195	75934	0.00e+00 **
gAgA	V2	ee	Rong2018	Fadeev2022	1.000	119993	12126	0.00e+00 **
gAgA	V2	ee	Jiao2019	Fadeev2022	1.000	119990	31328	0.00e+00 **

χ^2 uniform: 10% fractional uncertainty applied uniformly. χ^2 weighted: per-point uncertainty estimated from log-log curvature of constraint curve. p-value shown for weighted method. *** p < 0.001.

Pair 1/7 | gsgs · UNKNOWN · ee

Matter dataset	Delaunay2017	Antimatter dataset	Adkins2022
Interaction class	lepton-lepton	Lambda range	8.17e-13 – 1.88e-09 m
Matter unit	m	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	0.8727	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	93298.8	dof = 300 *** (p < 0.001)
chi2 (weighted)	10771.6	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.115	< 1 means weighted is more conservative
Mean sigma_matter	9.1%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	25.5%	Per-point uncertainty from curve curvature
lambda min	8.169e-13 m	
lambda max	1.879e-09 m	

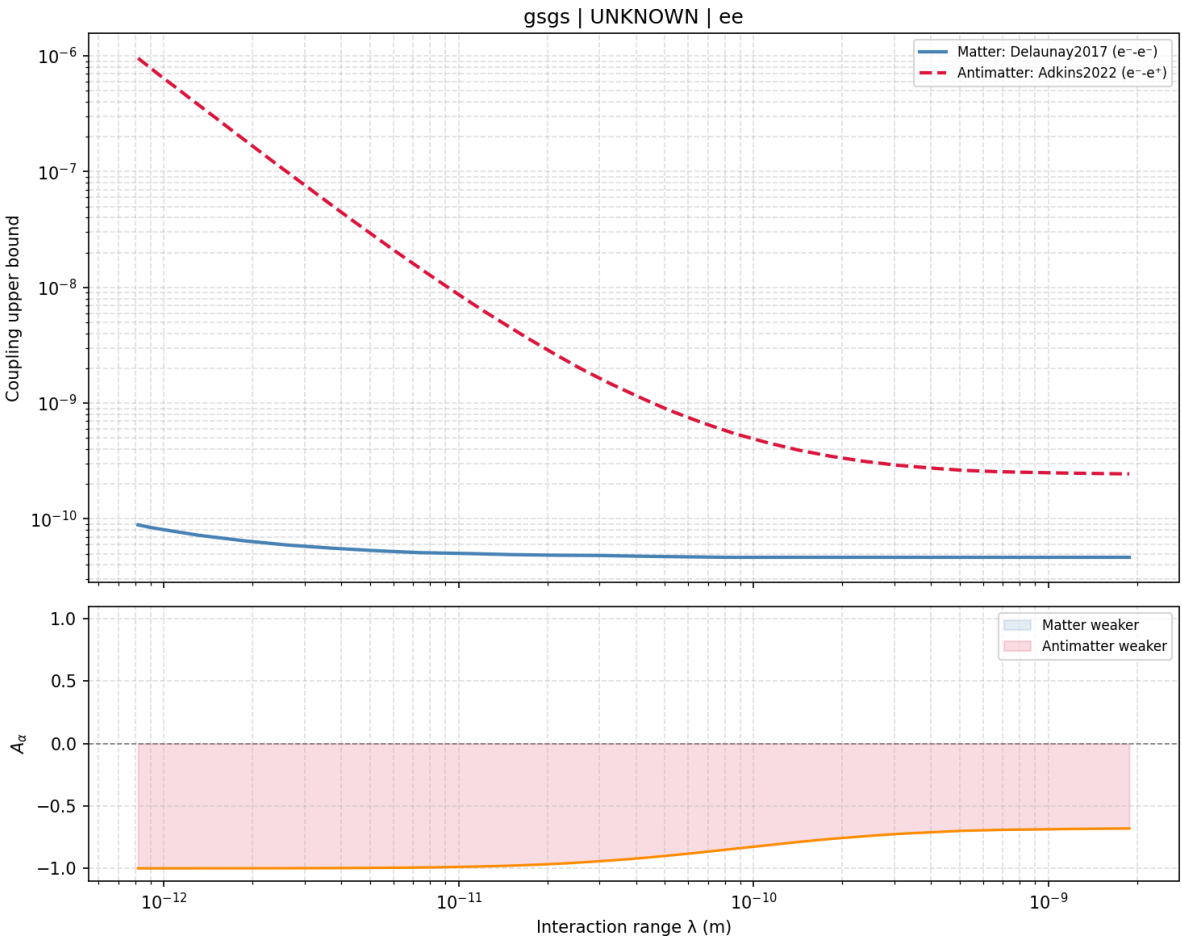


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under UNKNOWN potential. Shaded region shows combined uncertainty band from weighted analysis.

Pair 2/7 | gAgA · V1 · ep

Matter dataset	Karshenboim2011	Antimatter dataset	Fadeev2022
Interaction class	lepton-nucleon	Lambda range	2.44e-11 – 1.93e-08 m
Matter unit	m (pre-converted)	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	0.9998	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	119956.2	dof = 300 *** (p < 0.001)
chi2 (weighted)	101908.6	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.850	< 1 means weighted is more conservative
Mean sigma_matter	15.3%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	6.9%	Per-point uncertainty from curve curvature
lambda min	2.439e-11 m	
lambda max	1.929e-08 m	

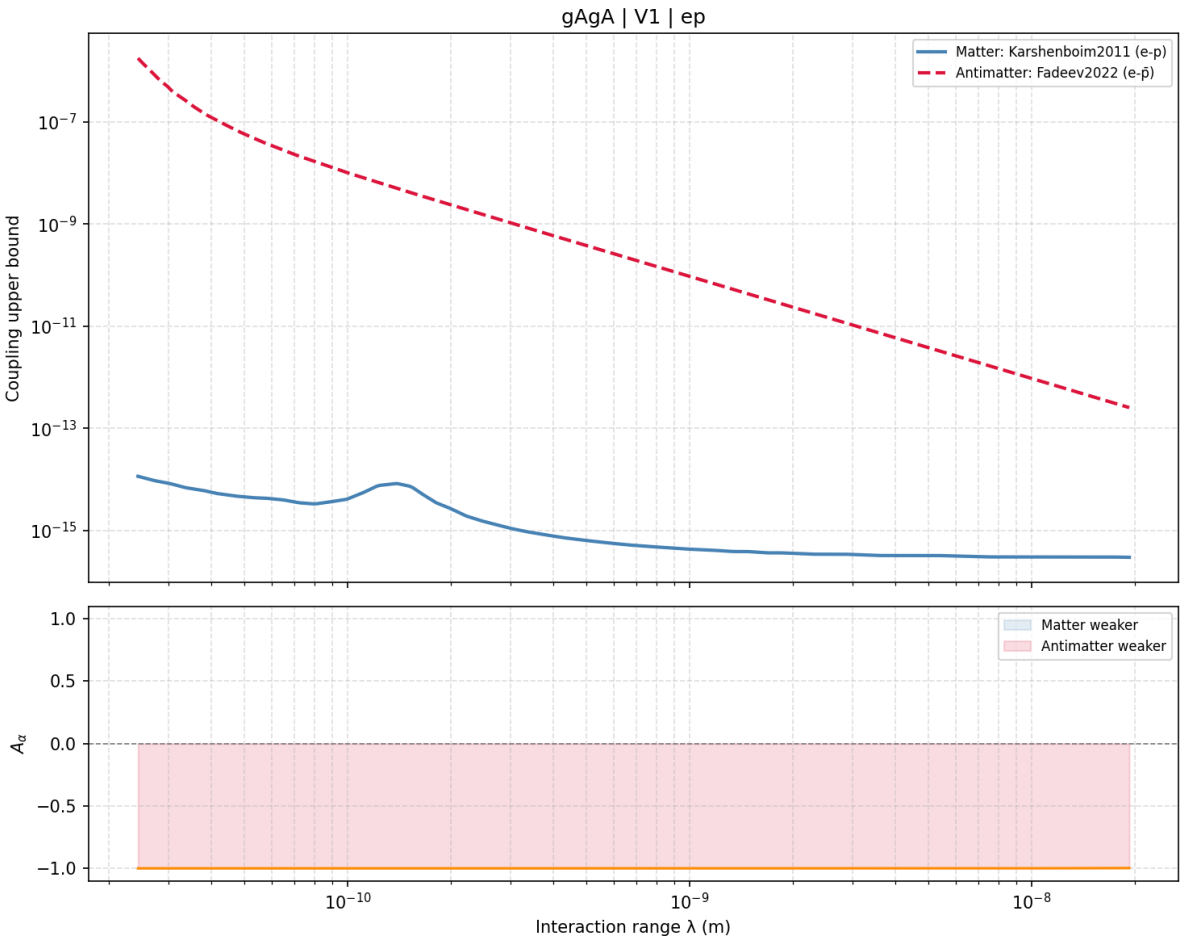


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ep sector under V1 potential. Shaded region shows combined uncertainty band from weighted analysis.

Pair 3/7 | gAgA · V2 · ee

Matter dataset	Kotler2015	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	1.75e-07 – 1.63e-05 m
Matter unit	m	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	1.0000	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	119989.9	dof = 300 *** (p < 0.001)
chi2 (weighted)	33391.2	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.278	< 1 means weighted is more conservative
Mean sigma_matter	16.6%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	5.0%	Per-point uncertainty from curve curvature
lambda min	1.754e-07 m	
lambda max	1.629e-05 m	

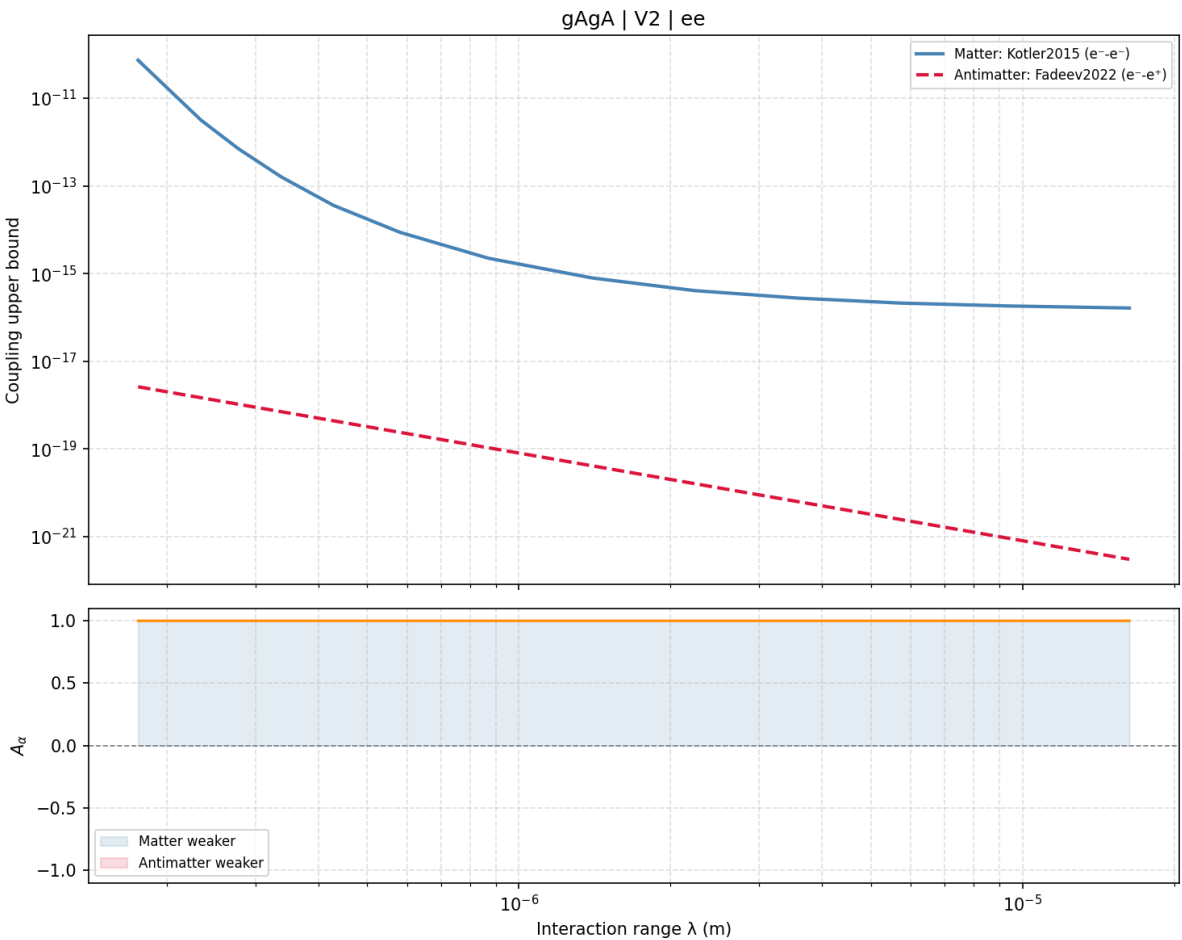


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential. Shaded region shows combined uncertainty band from weighted analysis.

Pair 4/7 | gAgA · V2 · ee

Matter dataset	Ficek2017	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	1.97e-11 – 1.31e-05 m
Matter unit	m	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	0.9990	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	119758.9	dof = 300 *** (p < 0.001)
chi2 (weighted)	65969.4	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.551	< 1 means weighted is more conservative
Mean sigma_matter	10.5%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	12.0%	Per-point uncertainty from curve curvature
lambda min	1.970e-11 m	
lambda max	1.310e-05 m	

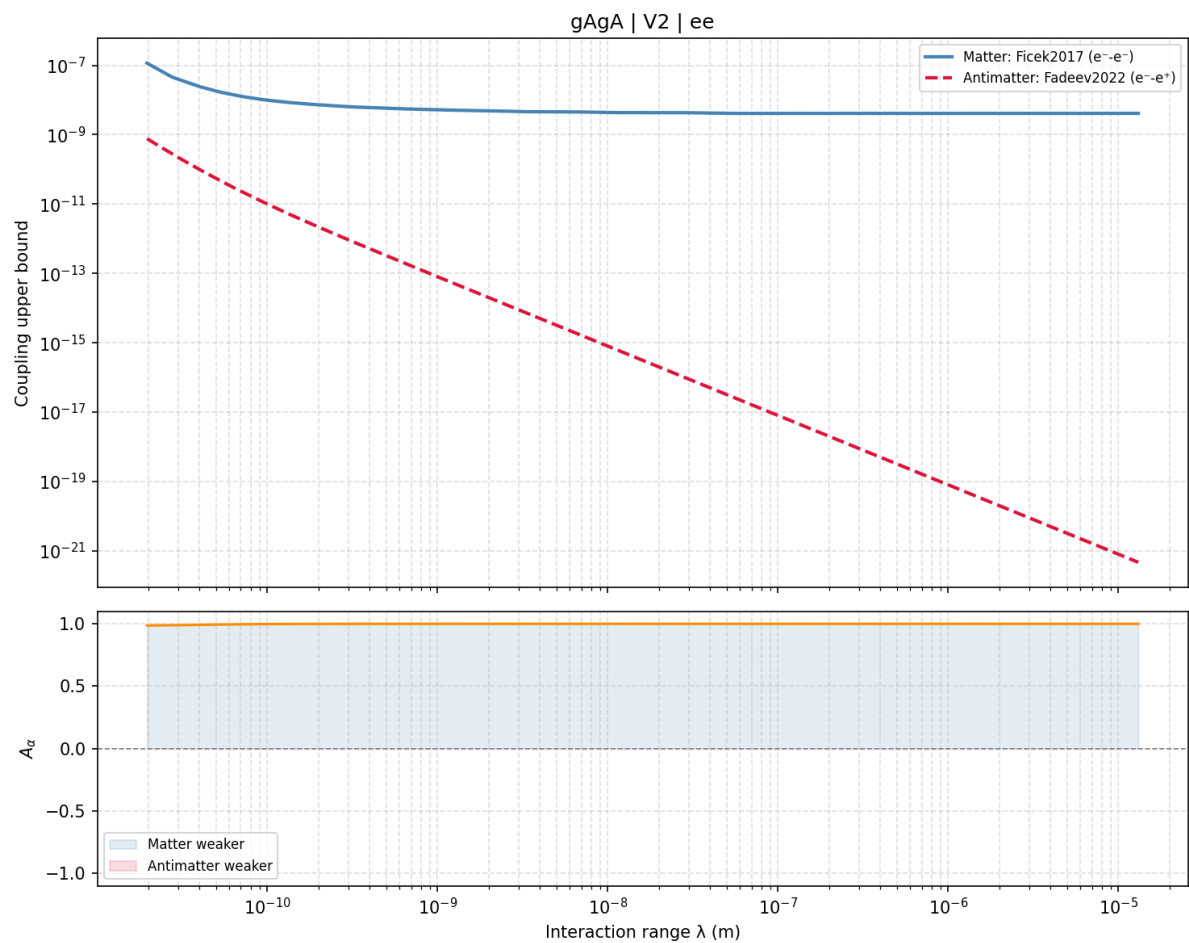


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential. Shaded region shows combined uncertainty band from weighted analysis.

Pair 5/7 | gAgA · V2 · ee

Matter dataset	Ficek2017	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	2.50e-12 – 1.63e-05 m
Matter unit	m	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	0.9539	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	109194.9	dof = 300 *** (p < 0.001)
chi2 (weighted)	75934.3	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.695	< 1 means weighted is more conservative
Mean sigma_matter	15.3%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	13.6%	Per-point uncertainty from curve curvature
lambda min	2.505e-12 m	
lambda max	1.629e-05 m	

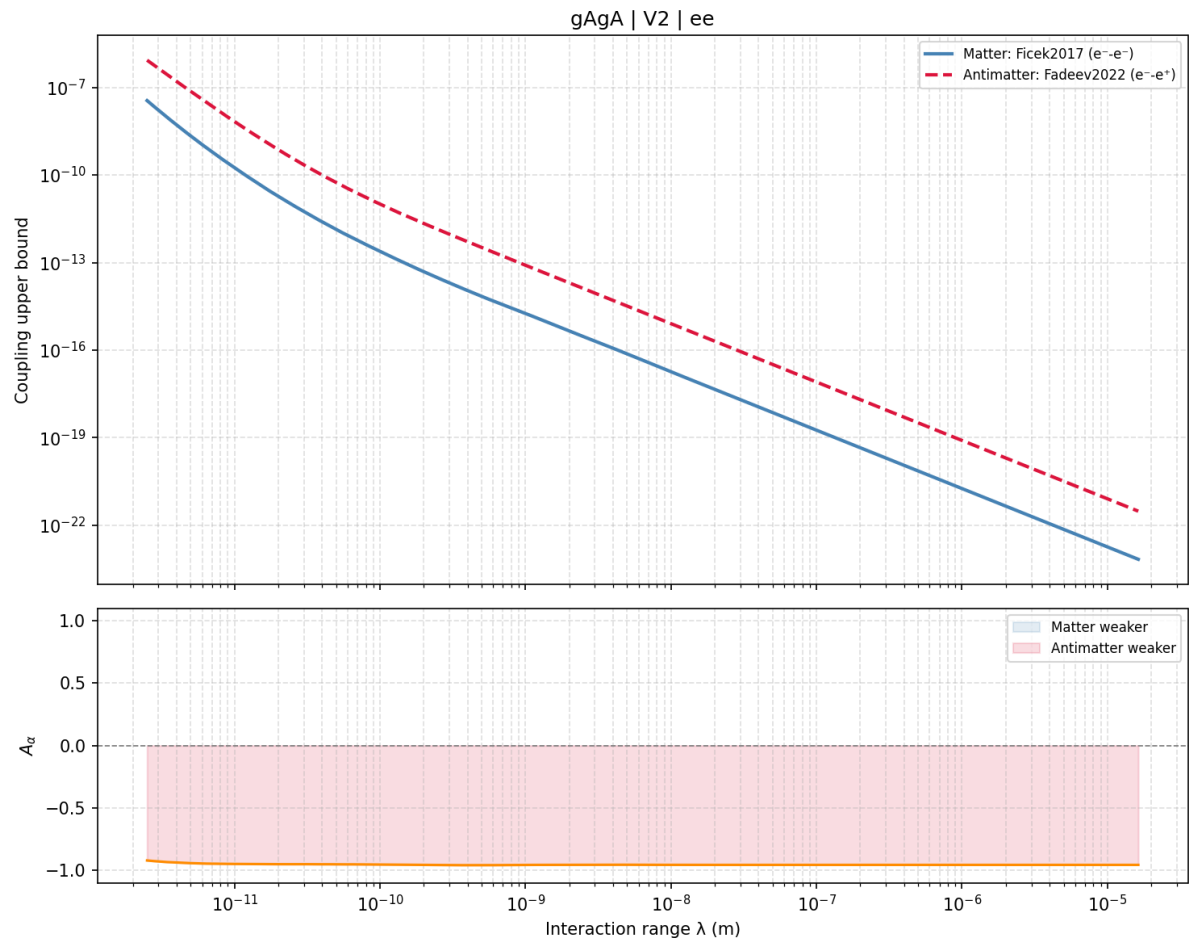


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential. Shaded region shows combined uncertainty band from weighted analysis.

Pair 6/7 | gAgA · V2 · ee

Matter dataset	Rong2018	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	4.06e-06 – 1.63e-05 m
Matter unit	m	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	1.0000	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	119992.9	dof = 300 *** (p < 0.001)
chi2 (weighted)	12125.7	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.101	< 1 means weighted is more conservative
Mean sigma_matter	22.1%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	5.1%	Per-point uncertainty from curve curvature
lambda min	4.063e-06 m	
lambda max	1.629e-05 m	

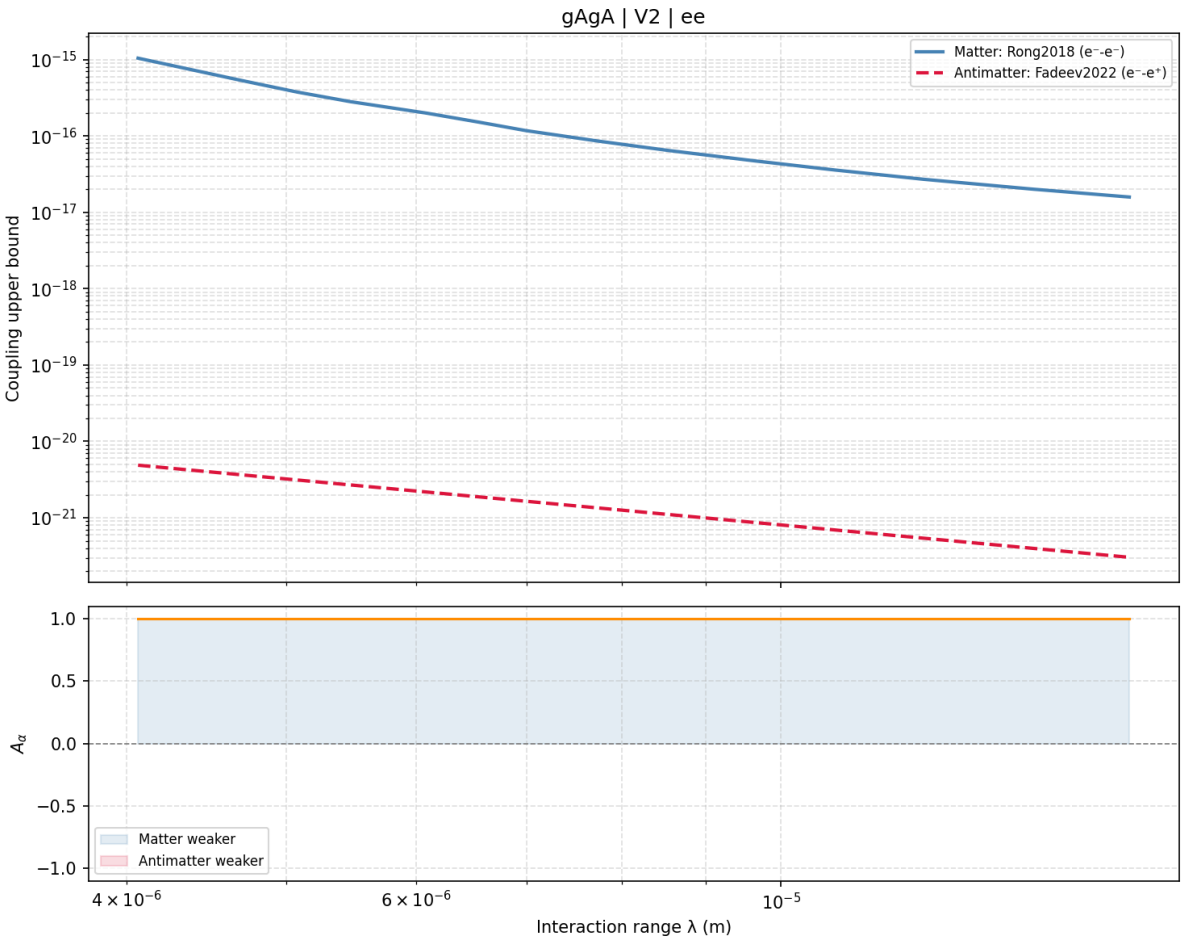


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential. Shaded region shows combined uncertainty band from weighted analysis.

Pair 7/7 | gAgA · V2 · ee

Matter dataset	Jiao2019	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	3.24e-09 – 3.83e-07 m
Matter unit	m	Antimatter unit	m

Statistic	Value	Interpretation
Mean A_alpha	1.0000	Strong asymmetry — large CPT-sensitive difference.
chi2 (uniform 10%)	119989.9	dof = 300 *** (p < 0.001)
chi2 (weighted)	31328.1	dof = 300 *** (p < 0.001)
p-value (uniform)	0.0000e+00	
p-value (weighted)	0.0000e+00	Preferred for thesis
chi2 ratio (w/u)	0.261	< 1 means weighted is more conservative
Mean sigma_matter	15.4%	Per-point uncertainty from curve curvature
Mean sigma_antimatter	5.0%	Per-point uncertainty from curve curvature
lambda min	3.242e-09 m	
lambda max	3.825e-07 m	

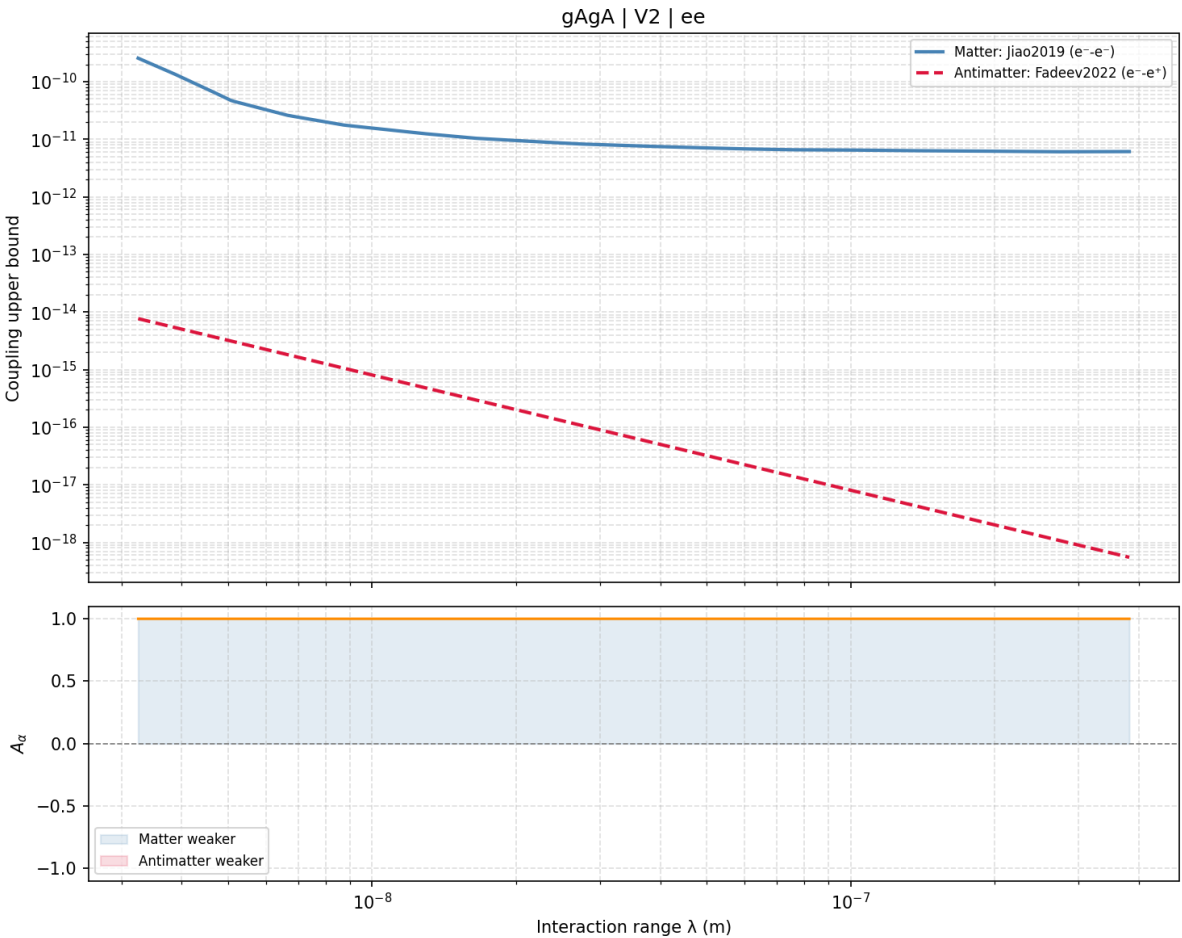


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential. Shaded region shows combined uncertainty band from weighted analysis.